Checkable memory for agents:

an audit of what an agent-built memory store can honestly claim

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Research paper · November 2026 Project repository: https://github.com/Anshul48/trace

Abstract

TRACE is a memory store for coding agents. It keeps a project's text as passages and answers a query by scoring only a small, routed shortlist of them instead of the whole store. Agents wrote most of TRACE's code and most of its evaluation reports, so we did not trust our own results. This paper describes the audit procedure we built to check them, and reports what survived. The procedure records whether each study had a plan frozen before it ran, ties every reported number to an evidence file by hash, and has a separate analyst recompute selected findings. One result held: guided routing scored 0.79 percent of rows on every shard (partition) and, averaged over 4 shards, kept 0.9985 of the exact top ten results (scoring work only, not latency; no plan file). Several did not: dense (meaning-based) search found fewer labelled sources than plain keyword search (26/50 vs 41/50); a hierarchy step and two keyword-plus-dense fusions did not beat keyword search; and the live guided service was far slower than a standard index at the search stage. When we seeded 18 faults into a copy of the paper, the automated check caught the mechanical ones and missed the semantic ones. These results concern one project's artifacts; downstream coding performance and the general effectiveness of the procedure were not evaluated.

1 What TRACE is, and why we audited it

TRACE is the memory layer of an agent system we are building. Its job is to let a coding agent find the right piece of a large body of project text (code, notes, reports) without reading all of it. The store studied here is sealed, meaning no rows were changed during the studies. It holds 812,545 passages, called units, in 4 disjoint partitions called shards. Each unit has a dense embedding, a vector that encodes its meaning. A query is answered in two steps: guided routing selects a shortlist of units, and only that shortlist is scored exactly against the query. When a unit matches, the service returns its parent text, the wider passage around it, so the agent receives context and not a fragment.

The longer-term design goal for TRACE is a memory that organises itself as it is used. Nothing in this paper tests that goal. This paper asks a narrower question about the store as it exists today: what can we honestly say it does?

The question matters because of how TRACE was built. Coding agents wrote the code, ran the experiments and wrote the reports. Generated code and generated reports can agree with each other while both evaluate the wrong scope; agreement alone does not show useful work. Early in the project a summary quoted a keyword-search score that belonged to a different experimental arm (see Corrections and limits). We needed a way to check each claim against evidence, and to keep corrections and failures on the record.

2 The audit procedure

The procedure works at the level of artifacts (files) and has four parts.

Plan status. Each study is labelled by when its plan was fixed: prospectively frozen (a preregistration file stating question, protocol, metrics and verdict rule, hashed before the run), historical (measured with no attested plan file), or specified after inspecting data. The hash is recorded in a local file. It shows that the plan

file still matches the recorded digest, provided that local record is untampered, so freezes are self-attested only.

Receipts. Every result written in digits in the prose carries a receipt: the file it came from, the hash of that file, and the line or key where the value sits. A build step, which we call the gate, fails when a digit-form number it recognises (outside its exemptions) has no receipt, when a pinned file has changed, or when the printed value differs from the receipted one. Numbers spelled out in words are invisible to the gate, so wording still needs review.

Separate recomputation. A second analyst recomputed selected results from raw files: the head-to-head verdicts, the rank-fusion and hierarchy figures, and the oracle-gap study. Results recomputed this way are marked OBSERVED. Results backed only by a hash-pinned summary are marked REPORTED; the score-fusion count, for example, was taken from the worker model (an AI agent) that ran it. Only selected results were recomputed, and recomputation checks values, not meaning.

Wording limit. A claims fence restricts what the text may say about each result (allowed-wording limit; 'prove' is disallowed), and a claim-to-evidence trail ships alongside the paper.

We also ran a control on our own acceptance tests. When an agent rewrites a program, we accept the rewrite if it agrees with the original on sampled inputs. In a prospectively specified control at the largest declared input domain, such gates passed while the programs differed in 200 of 200 seeds. We therefore call these accept loops fuzz-gated only: agreement on sampled inputs, not sameness. This matches the closed-form pass probability; it is not a discovery about real gates. The figure shows the whole flow with three examples from this project.

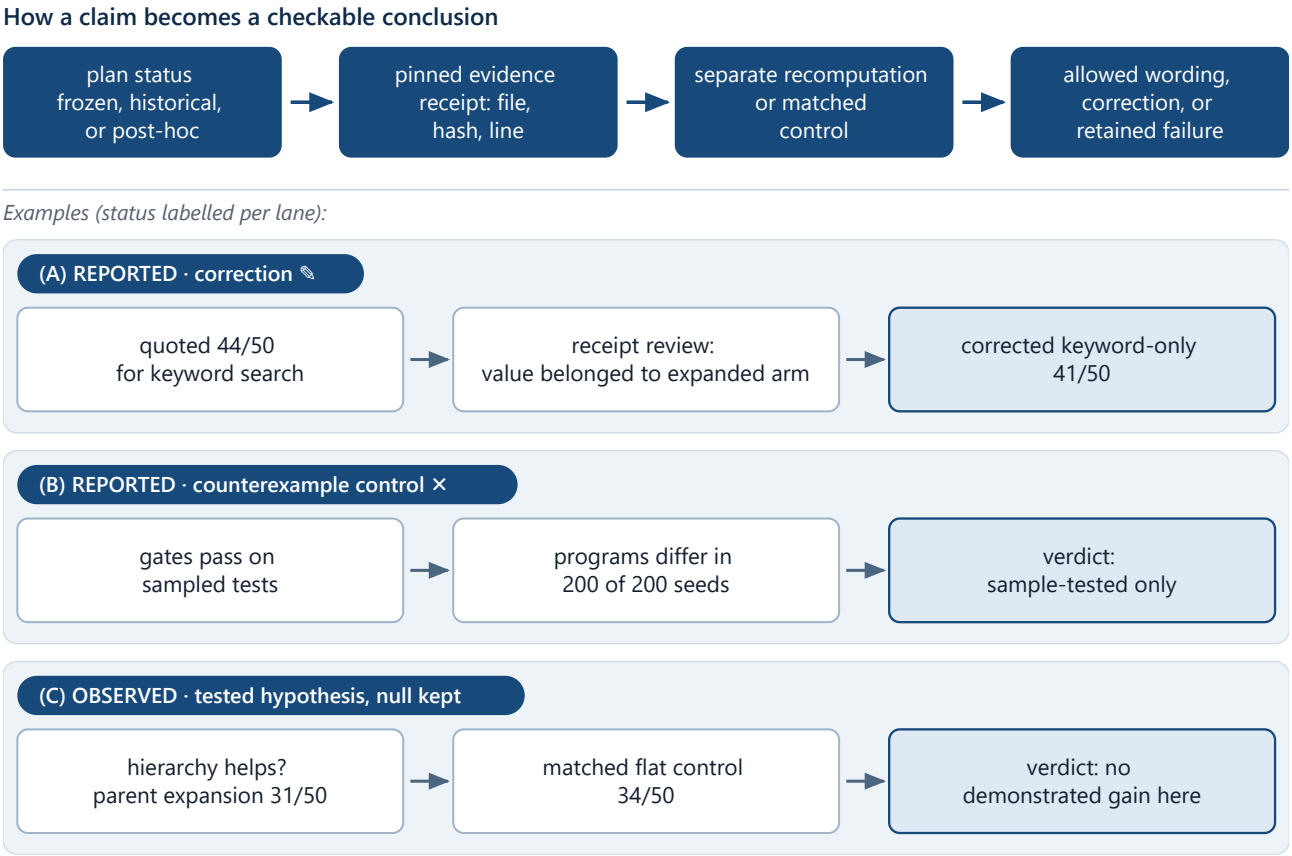


Figure 1. Claim flow: plan status, pinned evidence, separate recomputation or control, then allowed wording, correction, or retained failure. Lane A is a historical correction: keyword search quoted at 44/50 was the expanded-context arm, corrected to 41/50 for keyword search alone. Lane B is the control described above: gates passed while programs differed in 200 of 200 seeds, so such accept loops are sample-tested only. Lane C is a tested hypothesis kept as a null: parent expansion 31/50 against matched flat 34/50, no demonstrated gain here.

3 What the gate catches, and what it misses

To test the gate we seeded 18 faults into copies of an earlier snapshot of this paper (85 receipted numeric spans) and ran the gate on each copy, with 6/6 harmless control edits correctly passing. The gate caught 10 faults and missed 8. Every caught fault was mechanical: a digit-form number with no receipt, a changed pinned file, a span that differs from its receipted value, or a banned phrase. Every miss was semantic: a correct numeral attached to the wrong arm or denominator, a claim of prospective registration for a study whose table row says no plan file, a word-form number contradicting a receipted one, and a receipt id reused for a different quantity.

We chose the fault set ourselves, so it gives no error rate and says nothing about human reviewers. It shows that the gate checks values and hashes, and that meaning needs a reader.

4 What survived the audit: results on the TRACE store

Unless stated otherwise, results use 50 test searches over the sealed store. A search scores a source hit when the exact passage the query was labelled with appears among the returned results. Source hit measures finding a known source; it does not measure answer quality.

Table 1. Claim table for the main questions. OBSERVED means recomputed from raw files, some by the second analyst; REPORTED means a summary-backed figure we did not recompute. The preservation figure is a mean over 200 query-by-shard pairs (queries repeat across shards), and the scoring fraction holds on every shard.

Question and record level	Recorded result	What it allows us to say	Receipts
Does guided routing preserve the dense ranking while scoring fewer rows? OBSERVED (no plan file)	Exact top-ten overlap 0.9985; scored fraction 0.79 percent on every shard	Routing preserved this retriever's results at a small fraction of the scoring work. No inference about latency or usefulness	`e1a-recall-09985`, `e1a-rows-079`
Does dense search find the labelled source more often than keyword search? REPORTED, later reproduced as OBSERVED (frozen plan)	Keyword 41/50; dense 26/50	No. Keyword search found more labelled sources on these searches	`e2d-bm25-41-50`, `e2d-dense-26-50`
Does hierarchy help at matched (same-size) context? OBSERVED (frozen plan)	Expanded 31/50; flat 34/50	No demonstrated hierarchy gain; no equivalence claim	`e3a-e-hnsw-31-50`, `e3a-fk-hnsw-34-50`
Does combining keyword and dense scores beat keyword alone? OBSERVED (rank fusion, recomputed) and REPORTED (score-fusion count, taken from the worker) (frozen plan)	Rank fusion 38/50; score fusion 40/50; keyword 41/50	No gain over keyword in these fixed configurations; not a universal rejection of hybrids	`e3b-rrf-38-50`, `e3b-minmax-40-50`, `e3b-bm25-41-50`
Could an ideal chooser between keyword and dense do better than dense alone? OBSERVED (frozen plan)	Oracle gap +0.068 over dense in equal-style-average source-hit rate	The two methods find different sources on this constructed set; an oracle uses labels and is not a deployed router	`x1-gap`
Does fixed score fusion help inside the rows that routing admits? OBSERVED (stage 1 had no plan file; stages 2 to 6 were frozen before their runs, per our own freeze records)	Fusion cleared its source-hit bar on 3 of the first 5 query sets and missed on 2	A mixed pattern (post-hoc tally, no joint test) with no established explanation	`cf2-dhit`, `cf3-dhit`, `cf4-dhit`, `cf5-m-dhit`, `cf5-h-dhit`

A hash-pinned summary is weaker than a raw-data recomputation; the separate analyst recomputed rank fusion but took score-fusion counts from the worker.

Routing does less scoring work and keeps the ranking. Guided routing scored 0.79 percent of rows on every shard and kept 0.9985 of the exact top ten, measured with 50 test searches keeping the top 10 per shard. This is a measure of scoring work, not of latency or usefulness, and the measured TRACE arm used

routed dense search only, with no keyword component.

Dense search finds fewer labelled sources than keyword search. Keyword (word-match) search found the labelled source in 41/50 searches and dense search in 26/50. A later frozen-protocol re-collection of the same searches by an independent evaluator reproduced both counts: keyword 41 of 50, routed dense search 26 of 50.

Hierarchy and simple fusion did not help. Returning hierarchy-expanded parent context reached 31/50 against a flat arm of matched size at 34/50. Two ways of combining keyword and dense results, rank fusion and score fusion, reached 38/50 and 40/50 against keyword search alone at 41/50. These failures are kept at the same record level as the positive result.

Keyword and dense search are complementary on a constructed set. On frozen synthetic queries with strict source-unit labels, an oracle that picks the better of keyword and dense search gains +0.068 in equal-style-average source-hit rate over dense search alone (averaged equally over three query styles: question, keyword, description). The labels credit only the passage a query was written from, so this is complementarity on a constructed set, not a deployed router. The evaluation uses unrestricted retrieval, not routed TRACE, and a single embedding model (MiniLM-L6-v2) on one corpus. On the original fifty searches an oracle between keyword and dense gains only +2/50 over keyword search (REPORTED), so the two oracle gains are not comparable.

Fusion inside the route gave a mixed pattern. The route is the unchanged guided search path; its candidate set is the rows it admits for a query, on average 13.4% of the store (wider than the scored shortlist). Stage 1 had no plan file: the unchanged route admits the labelled source for 45/50 of the original searches. In stages 2 to 6, each frozen before its run per our own freeze records, a fixed, untrained equal-weight fusion of keyword and dense scores was compared with routed dense search on synthetic model-written queries. A query set clears when fusion's source-hit gain is at least +0.04, its reciprocal-rank gain at least +0.03, and both lower document-bootstrap interval bounds exceed 0.

Stage 2 (388 queries): source-hit +0.055 over dense search; cleared. Stage 3 (300 fresh queries): +0.033 (interval [0.001,0.066]), below the source-hit bar, so not confirmed. Stage 4 (297 queries written by a small Claude-family model): +0.087 (interval [0.050,0.125]); cleared. Stage 5 (300 fresh passages, two authors): the worker-written set gave +0.023 (interval [0.003,0.044]), below the bar; the Claude-family set gave +0.053 (interval [0.019,0.090]), clearing it. As a post-hoc tally with no joint test, fusion cleared its source-hit bar over dense search on 3 of the first 5 sets and missed on 2. Author identity alone does not explain the outcomes: the paired author contrast on the same passages was +0.030 [-0.009,0.071], an interval that includes zero.

Stage 6 tested word overlap directly: the worker models wrote queries for the same 300 fresh passages twice, avoiding the passage's words (mean word overlap 0.216) or reusing them (0.777). The preregistered primary (the paired difference in fusion's source-hit gain between conditions) needed +0.03 and was +0.000 [-0.050,0.050]. The interval includes both zero and the bar, so an overlap effect is neither shown nor excluded. Why the gains varied remains unestablished, and nothing here licenses a deployed hybrid.

Speed: a standard index is far faster at the search stage. In the same-store head-to-head, the fastest quality-matched (same answer quality) standard-index configuration recorded a median of 0.66 milliseconds (search only, 1 thread, ids only, shared virtual machine) against 1236 milliseconds for the live guided service, a ratio of 1884 (from unrounded medians). The preregistered verdict reads that guided search is dominated by a standard index on this store. Embedding on both sides leaves a ratio near 86. This compares search-stage cost only: the service returns parent text, the standard index ids only. Waiting on shard-engine processes, which run in sequence, took 88.8 to 91.2 percent of guided query time in a profile (REPORTED); launch and scoring were not separated. We make no general speed claim.

Memory: the preregistered advantage criterion failed. In the memory-at-scale comparison, the ratio of TRACE's process memory to a standard graph index's was 0.1675 at small size, 0.1717 at middle size, and 0.1721 at a third, largest store size (the smaller size is a worker figure we did not re-time; settings differed between sizes). The ratio did not fall with scale, so it failed the preregistered memory-advantage criterion. This process memory omits the page cache, and we draw no memory claim.

5 Corrections and limits

One correction shows the procedure narrowing a claim. An earlier note quoted 44/50 for keyword search, which was the expanded-context arm; the keyword-only figure is 41/50. The error was found by reading receipts, not through the automated gate.

Queries were authored, not sampled from user traffic, and one model family both wrote and plausibility-filtered the first frozen synthetic set, so author and judge effects were never separated; no synthetic query was written by a person. A different-model stand-in transfer check on three small public sets did not reproduce the headline recall and scoring work (REPORTED, words only), so the scoring-work figure may not transfer. Source hit counts only the exact source passage, and other passages that answer are missed. The sealed-store text-retrieval studies use one corpus and a single embedding model (the memory comparison used another store), and 50 test searches give coarse resolution. No general speed claim and no memory claim are made for TRACE against a standard index.

6 Reproduction

An experiment bundle re-derives the small-experiment results from bundled scripts and data with one command (reruns on the author's host reproduced the shipped verdicts, with three analyses byte-identical; other systems are untested). The sealed-store runs need the author's virtual machine and sealed index and cannot be re-derived from the bundle alone; a receipt bundle holds the pinned result files, and the checker confirms that each recognised numeric span matches its pinned file. It confirms values and hashes only; a reader should use the claim table.

7 What we take from this

For TRACE, the audit leaves one positive result standing: routing cuts scoring work sharply while preserving the dense ranking. It also leaves a plain list of things the current store does not do better than simple baselines: finding labelled sources, search-stage speed, and a memory advantage that grows with scale. Those are the measured starting point for the next version, not a verdict on the design goal, which this paper did not test.

For agent-built software more generally, our experience is that a mechanical gate is worth having because it is cheap and catches mechanical errors, and that it does not replace a reader: every semantic fault we planted passed it. Keeping the failures at the same record level as the successes was what made the one success credible to us.

> **AI-ASSISTANCE DISCLOSURE**
> Built with AI coding agents (Muse Code, Claude, Codex) under human architectural direction. Digit-form numbers are checked at build time against hash-pinned result files (some of them summaries); findings stated in words are not receipted.

Appendix A Receipted evidence

One row per receipt id cited by the Writeup, in order of first appearance. Evidence paths under owner-hub are shown relative to owner-hub/ (full path in the manifest and in each cell tooltip). CURRENT sha256 is the evidence file hash at build time (2026-11-12T20:09:42Z).

#	id	displayed value	tag	evidence file	line or key	recorded sha256	current sha256	status
1	e1a-rows-079	0.79 cited 3x	OBSERVED	scratch/wave6/paper/evidence-vm/shard-fraction-all/per_shard.tsv	2	78fb3c4db2298d3e01b34dafc4ac55dec3f7e6a56b89d461638a3c9762ab9964	78fb3c4db2298d3e01b34dafc4ac55dec3f7e6a56b89d461638a3c9762ab9964	MATCH
2	num-shards-4	4 cited 2x	REPORTED	scratch/wave6/claude-sci/h2h/baselines-mirror/RESULTS.md	9	0d6f09952ac72d71960153562cd794f302ea176c0c9b1f9842683a9dafc24912	0d6f09952ac72d71960153562cd794f302ea176c0c9b1f9842683a9dafc24912	MATCH
3	e1a-recall-09985	0.9985 cited 3x	OBSERVED	scratch/wave6/paper/evidence-facts/FACTS-AND-RULES.md	23	6a4434c26ab20e7e6ffa2876b5e02fcec3c59e498342a731b74f4a57de7ef249	6a4434c26ab20e7e6ffa2876b5e02fcec3c59e498342a731b74f4a57de7ef249	MATCH
4	e2d-dense-26-50	26/50 cited 3x	REPORTED	scratch/wave6/claude-sci/hier/summary_hl.json	F10_hnsw.hits/F10_hnsw.n	c5c3fea0d7b9b6e3af6d757d3936b25695fcf1e0bdb3dbf0fe894dff96d81cb2	c5c3fea0d7b9b6e3af6d757d3936b25695fcf1e0bdb3dbf0fe894dff96d81cb2	MATCH
5	e2d-bm25-41-50	41/50 cited 3x	REPORTED	scratch/wave6/claude-sci/hier/RESULT.md	42	e577ef0f71d69ee6f687079bffce66441619f3ed633a677a1a2e21ecd9578e37	e577ef0f71d69ee6f687079bffce66441619f3ed633a677a1a2e21ecd9578e37	MATCH
6	fc-seeded-18	18 cited 2x	REPORTED	scratch/wave6/paper/evidence-fc/RESULT.md	Headline: Seeded faults: 18	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	MATCH
7	e1a-units-812545	812,545	REPORTED	scratch/wave6/paper/evidence-facts/FACTS-AND-RULES.md	22	6a4434c26ab20e7e6ffa2876b5e02fcec3c59e498342a731b74f4a57de7ef249	6a4434c26ab20e7e6ffa2876b5e02fcec3c59e498342a731b74f4a57de7ef249	MATCH
8	e2fz-200of200-2e20	200 of 200 cited 2x	REPORTED	scratch/wave6/claude-sci/e2/run.log	7	0597f54106d245c68479cd7809cb5b313a6cba68e997375a32621773136383e6	0597f54106d245c68479cd7809cb5b313a6cba68e997375a32621773136383e6	MATCH
9	l8-bm25-44-50	44/50 cited 2x	REPORTED	scratch/wave6/paper/positions/corrections.md	2	466a524fc5255db18fc02264a39f6895b4bf959715c005bec925956569f0a8a4	466a524fc5255db18fc02264a39f6895b4bf959715c005bec925956569f0a8a4	MATCH
10	l8-receipt-41-50	41/50 cited 2x	REPORTED	scratch/wave6/paper/positions/corrections.md	2	466a524fc5255db18fc02264a39f6895b4bf959715c005bec925956569f0a8a4	466a524fc5255db18fc02264a39f6895b4bf959715c005bec925956569f0a8a4	MATCH
11	e3a-e-hnsw-31-50	31/50 cited 3x	OBSERVED	scratch/wave6/claude-sci/hier/RESULT.md	40	e577ef0f71d69ee6f687079bffce66441619f3ed633a677a1a2e21ecd9578e37	e577ef0f71d69ee6f687079bffce66441619f3ed633a677a1a2e21ecd9578e37	MATCH
12	e3a-fk-hnsw-34-50	34/50 cited 3x	OBSERVED	scratch/wave6/claude-sci/hier/RESULT.md	40	e577ef0f71d69ee6f687079bffce66441619f3ed633a677a1a2e21ecd9578e37	e577ef0f71d69ee6f687079bffce66441619f3ed633a677a1a2e21ecd9578e37	MATCH
13	fc-snapshot-85	85	REPORTED	scratch/wave6/paper/evidence-fc/RESULT.md	11	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	MATCH
14	fc-controls-6	6/6	REPORTED	scratch/wave6/paper/evidence-fc/RESULT.md	Controls 6/6 pass	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	MATCH
15	fc-caught-10	10	REPORTED	scratch/wave6/paper/evidence-fc/RESULT.md	Headline: Checker caught 10	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	MATCH
16	fc-missed-8	8	REPORTED	scratch/wave6/paper/evidence-fc/RESULT.md	Headline: missed 8	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	bcee2ab877f478ffb32cfd7eaf89ab336c70035cefd05271ac27594becaf7477	MATCH
17	e1a-n50	50 cited 5x	REPORTED	scratch/wave6/paper/evidence-facts/FACTS-AND-RULES.md	23	6a4434c26ab20e7e6ffa2876b5e02fcec3c59e498342a731b74f4a57de7ef249	6a4434c26ab20e7e6ffa2876b5e02fcec3c59e498342a731b74f4a57de7ef249	MATCH
18	mfid-n200	200	REPORTED	scratch/wave6/claude-sci/h2h/baselines-mirror/RESULTS.md	23	0d6f09952ac72d71960153562cd794f302ea176c0c9b1f9842683a9dafc24912	0d6f09952ac72d71960153562cd794f302ea176c0c9b1f9842683a9dafc24912	MATCH
19	e3b-rrf-38-50	38/50 cited 2x	OBSERVED	scratch/wave6/claude-sci/hier/HIER-FINDING.md	11	d3af34951771c1789833b4dbeed3996bfccf238a3c870e7e689651c08c6805e4	d3af34951771c1789833b4dbeed3996bfccf238a3c870e7e689651c08c6805e4	MATCH

#	id	displayed value	tag	evidence file	line or key	recorded sha256	current sha256	status
20	e3b-minmax-40-50	40/50 cited 2x	REPORTED	scratch/wave6/claude-sci/hier/HIER-FINDING.md	11	d3af34951771c1789833b4dbeed3996bfccf238a3c870e7e689651c08c6805e4	d3af34951771c1789833b4dbeed3996bfccf238a3c870e7e689651c08c6805e4	MATCH
21	e3b-bm25-41-50	41/50 cited 2x	OBSERVED	scratch/wave6/claude-sci/hier/HIER-FINDING.md	11	d3af34951771c1789833b4dbeed3996bfccf238a3c870e7e689651c08c6805e4	d3af34951771c1789833b4dbeed3996bfccf238a3c870e7e689651c08c6805e4	MATCH
22	x1-gap	+0.068 cited 2x	OBSERVED	scratch/wave6/claude-sci/x1/vm/RESULT.md	19	492c12c17d22a0d013abef7033703f3d95903f51746a4ccb3e3efa29dd8aad6d	492c12c17d22a0d013abef7033703f3d95903f51746a4ccb3e3efa29dd8aad6d	MATCH
23	dt-five-cleared-3	3 cited 2x	REPORTED	scratch/wave6/paper/evidence-derived/tally-five-sets.txt	12	bec3c4376e98b750ec12aefbff3583a1e60cf29934a8efc6c0b176cb9eed6a99	bec3c4376e98b750ec12aefbff3583a1e60cf29934a8efc6c0b176cb9eed6a99	MATCH
24	num-five-sets-5	5 cited 2x	REPORTED	scratch/wave6/paper/evidence-cfuse/c-fuse-5/OWN-CHECK.md	6	5d2812c93b2b1a43ba4c681e3e35e1d61f6d35dc63d66fee937bda507a925bce	5d2812c93b2b1a43ba4c681e3e35e1d61f6d35dc63d66fee937bda507a925bce	MATCH
25	dt-five-missed-2	2 cited 2x	REPORTED	scratch/wave6/paper/evidence-derived/tally-five-sets.txt	12	bec3c4376e98b750ec12aefbff3583a1e60cf29934a8efc6c0b176cb9eed6a99	bec3c4376e98b750ec12aefbff3583a1e60cf29934a8efc6c0b176cb9eed6a99	MATCH
26	e1a-k10	10	REPORTED	scratch/wave6/paper/evidence-facts/FACTS-AND-RULES.md	23	6a4434c26ab20e7e6ffa2876b5e02fcec3c59e498342a731b74f4a57de7ef249	6a4434c26ab20e7e6ffa2876b5e02fcec3c59e498342a731b74f4a57de7ef249	MATCH
27	sealed50-k-41	41	OBSERVED	scratch/wave2/sealed-eval-stage2b-v1/metrics-report.txt	17	092da4856d110d02743dc044404112c1d8189472a37d6a928fd41c8b8d917543	092da4856d110d02743dc044404112c1d8189472a37d6a928fd41c8b8d917543	MATCH
28	sealed50-g-26	26	OBSERVED	scratch/wave2/sealed-eval-stage2b-v1/metrics-report.txt	18	092da4856d110d02743dc044404112c1d8189472a37d6a928fd41c8b8d917543	092da4856d110d02743dc044404112c1d8189472a37d6a928fd41c8b8d917543	MATCH
29	r21-x0-styles	three	REPORTED	scratch/wave6/paper/evidence-derived/runner21/query-split.json	style_count_word	a6bcb3c04b568897a0b757114d1445255545e88eaf21bccd7d7e2eda6adce093	a6bcb3c04b568897a0b757114d1445255545e88eaf21bccd7d7e2eda6adce093	MATCH
30	old50-oracle-gain	+2/50	REPORTED	scratch/wave6/paper/positions/corrections.md	3	466a524fc5255db18fc02264a39f6895b4bf959715c005bec925956569f0a8a4	466a524fc5255db18fc02264a39f6895b4bf959715c005bec925956569f0a8a4	MATCH
31	cf2-cand-pct	13.4%	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-2.result.md	60	571cd513aedd44609b4073bf67ba6d32d29a66c12a6c1194966ae26ef2b251f4	571cd513aedd44609b4073bf67ba6d32d29a66c12a6c1194966ae26ef2b251f4	MATCH
32	cf1-admit-45	45/50	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-1.result.md	Ceilings: old 50 admission	8547c4ecfdf790fae6527a90048c3ccc3f47065da26c71cbf6575390b51e9576	8547c4ecfdf790fae6527a90048c3ccc3f47065da26c71cbf6575390b51e9576	MATCH
33	bar-hit	+0.04	VERIFIED	scratch/wave6/claude-sci/x3/PREREG15-CFUSE-S6.md	17	46c489c6c156a7c9518022a7cbcb29d6bcc4e3625bd7b0ca8bac346b13ecc740	46c489c6c156a7c9518022a7cbcb29d6bcc4e3625bd7b0ca8bac346b13ecc740	MATCH
34	bar-mrr	+0.03	VERIFIED	scratch/wave6/claude-sci/x3/PREREG15-CFUSE-S6.md	17	46c489c6c156a7c9518022a7cbcb29d6bcc4e3625bd7b0ca8bac346b13ecc740	46c489c6c156a7c9518022a7cbcb29d6bcc4e3625bd7b0ca8bac346b13ecc740	MATCH
35	num-ci-zero-0	0	REPORTED	scratch/wave6/claude-sci/x3/PREREG15-CFUSE-S6.md	17	46c489c6c156a7c9518022a7cbcb29d6bcc4e3625bd7b0ca8bac346b13ecc740	46c489c6c156a7c9518022a7cbcb29d6bcc4e3625bd7b0ca8bac346b13ecc740	MATCH
36	cf2-n	388	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-2.result.md	Primary comparison n	571cd513aedd44609b4073bf67ba6d32d29a66c12a6c1194966ae26ef2b251f4	571cd513aedd44609b4073bf67ba6d32d29a66c12a6c1194966ae26ef2b251f4	MATCH
37	cf2-dhit	+0.055	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-2.result.md	Verdict d_hit	571cd513aedd44609b4073bf67ba6d32d29a66c12a6c1194966ae26ef2b251f4	571cd513aedd44609b4073bf67ba6d32d29a66c12a6c1194966ae26ef2b251f4	MATCH
38	cf3-n	300	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-3.result.md	Primary comparison n	8843f993f83384e9a9db46e3ce46a84f64e45d77445cdc6484cbceb73574ffa7	8843f993f83384e9a9db46e3ce46a84f64e45d77445cdc6484cbceb73574ffa7	MATCH
39	cf3-dhit	+0.033	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-3.result.md	Verdict d_hit	8843f993f83384e9a9db46e3ce46a84f64e45d77445cdc6484cbceb73574ffa7	8843f993f83384e9a9db46e3ce46a84f64e45d77445cdc6484cbceb73574ffa7	MATCH

#	id	displayed value	tag	evidence file	line or key	recorded sha256	current sha256	status
40	cf3-ci-hit	[0.001,0.066]	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-3.result.md	5	8843f993f83384e9a9db46e3ce46a84f64e45d77445cdc6484cbceb73574ffa7	8843f993f83384e9a9db46e3ce46a84f64e45d77445cdc6484cbceb73574ffa7	MATCH
41	cf4-n	297	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-4.result.md	7	506989417559e769428c86713255de3944481c6afa5302b5721e2bc124582329	506989417559e769428c86713255de3944481c6afa5302b5721e2bc124582329	MATCH
42	cf4-dhit	+0.087	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-4.result.md	5	506989417559e769428c86713255de3944481c6afa5302b5721e2bc124582329	506989417559e769428c86713255de3944481c6afa5302b5721e2bc124582329	MATCH
43	cf4-ci-hit	[0.050,0.125]	OBSERVED	scratch/wave6/paper/evidence-cfuse/c-fuse-4.result.md	5	506989417559e769428c86713255de3944481c6afa5302b5721e2bc124582329	506989417559e769428c86713255de3944481c6afa5302b5721e2bc124582329	MATCH
44	cf5-n	300	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-5/RESULT.md	20	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	MATCH
45	cf5-m-dhit	+0.023	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-5/RESULT.md	5	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	MATCH
46	cf5-m-ci	[0.003,0.044]	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-5/RESULT.md	5	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	MATCH
47	cf5-h-dhit	+0.053	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-5/RESULT.md	7	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	MATCH
48	cf5-h-ci-p1	[0.019,0.090]	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-5/RESULT.md	7	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	MATCH
49	cf5-paired	+0.030 [-0.009,0.071]	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-5/RESULT.md	93	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	a559f63031c12262e9898eeb1c614d332bdb6ea335cc120037c157c351fff3cb	MATCH
50	cf6-n	300	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-6/RESULT.md	37	dcf4f6465587fbb817eec60c680529d35deafa5b8045301441ae4967090ac113	dcf4f6465587fbb817eec60c680529d35deafa5b8045301441ae4967090ac113	MATCH
51	cf6-ovl-l	0.216	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-6/RESULT.md	106	dcf4f6465587fbb817eec60c680529d35deafa5b8045301441ae4967090ac113	dcf4f6465587fbb817eec60c680529d35deafa5b8045301441ae4967090ac113	MATCH
52	cf6-ovl-h	0.777	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-6/RESULT.md	107	dcf4f6465587fbb817eec60c680529d35deafa5b8045301441ae4967090ac113	dcf4f6465587fbb817eec60c680529d35deafa5b8045301441ae4967090ac113	MATCH
53	bar-d	+0.03	VERIFIED	scratch/wave6/claude-sci/x3/PREREG15-CFUSE-S6.md	18	46c489c6c156a7c9518022a7cbcb29d6bcc4e3625bd7b0ca8bac346b13ecc740	46c489c6c156a7c9518022a7cbcb29d6bcc4e3625bd7b0ca8bac346b13ecc740	MATCH
54	cf6-d	+0.000 [-0.050,0.050]	VERIFIED	scratch/wave6/paper/evidence-cfuse/c-fuse-6/RESULT.md	5	dcf4f6465587fbb817eec60c680529d35deafa5b8045301441ae4967090ac113	dcf4f6465587fbb817eec60c680529d35deafa5b8045301441ae4967090ac113	MATCH
55	e2a-p50-066	0.66	OBSERVED	scratch/wave6/claude-sci/h2h/H2H-FINDING.md	7	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	MATCH
56	num-thread-1	1	REPORTED	scratch/wave6/claude-sci/h2h/H2H-FINDING.md	7	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	MATCH
57	e2b-guided-1236	1236	OBSERVED	scratch/wave6/claude-sci/h2h/H2H-FINDING.md	7	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	MATCH
58	e2b-r-1884	1884	OBSERVED	scratch/wave6/claude-sci/h2h/H2H-FINDING.md	7	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	MATCH
59	e2b-rend2end-86	86	OBSERVED	scratch/wave6/claude-sci/h2h/H2H-FINDING.md	7	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	ec162f0d51819ce311d11e553b28e23bca9f6d5f363dc614914a78bc2e33c9cd	MATCH

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60	lat-engine-888	88.8	REPORTED	scratch/wave6/paper/evidence-vm/latency-profile-r2/vm-trace-latency-profile-v1-r2.result.md	13	a6d9344841a093f9f03c812004564770daf39800a4aeaaf68c945efdac09d14e	a6d9344841a093f9f03c812004564770daf39800a4aeaaf68c945efdac09d14e	MATCH
61	lat-engine-912	91.2	REPORTED	scratch/wave6/paper/evidence-vm/latency-profile-r2/vm-trace-latency-profile-v1-r2.result.md	13	a6d9344841a093f9f03c812004564770daf39800a4aeaaf68c945efdac09d14e	a6d9344841a093f9f03c812004564770daf39800a4aeaaf68c945efdac09d14e	MATCH
62	d3-ratio-0p8	0.1675	OBSERVED	scratch/wave6/paper/d3/p0/FINDING-D3-0p8M.md	9	73a4e594ea5946a0cd76bf294f0e23e2ae7634e1a53f727f88e448503622a2a7	73a4e594ea5946a0cd76bf294f0e23e2ae7634e1a53f727f88e448503622a2a7	MATCH
63	d3-ratio-3p2	0.1717	OBSERVED	scratch/wave6/paper/d3/p1/FINDING-D3-3p2M.md	Ratios per prereg	ee58eae16ef10a85e0ec01a034e0278638e3fc3e2854bc1ff5da2dd74318c9ad	ee58eae16ef10a85e0ec01a034e0278638e3fc3e2854bc1ff5da2dd74318c9ad	MATCH
64	d3-ratio-5p0	0.1721	OBSERVED	scratch/wave6/paper/evidence-vm/d3-5m-r5/d3-5p0m-v1-r5.result.md	173	a8d51d768757e87c86a2a4b18fe32f68d4cf3fd1979f391542901d59a8ef0632	a8d51d768757e87c86a2a4b18fe32f68d4cf3fd1979f391542901d59a8ef0632	MATCH
65	r21-identical-analyses	three	REPORTED	scratch/wave6/paper/evidence-derived/runner21/reproduction-tally.json	identical_analyses_word	a7d49239d4efe01758581e6549f860edaf9c642816a3f0842a6137ad3c13f032	a7d49239d4efe01758581e6549f860edaf9c642816a3f0842a6137ad3c13f032	MATCH

Provenance. Built 2026-11-12T20:09:42Z | writeup html sha256: ed501957f3c9e855a6de475077fb74939aa150512862698c6944442e452b7b07 | receipts json sha256: 46d2da1150430a0272984a34628786091a55a00d92e4fd2e4129631d0c847089 | checker exit=0: OK: scratch/wave6/paper-pro-20261112/v2/writeup-v2.html (93 receipted spans, 247 text segments clean) | 65 cited receipts: 65 MATCH, 0 MISMATCH, 0 MISSING, 0 UNKNOWN; 0 known-lost uncited (MISSING)